

IMPACT OF COOLING CHANNEL SHAPE ON THE PERFORMANCE OF HOLLOW HAIRPIN WINDINGS

Electromagnetic–thermal analysis and multi-objective shape optimization of direct oil-cooled hairpin windings for traction motors

RESEARCH TEAM

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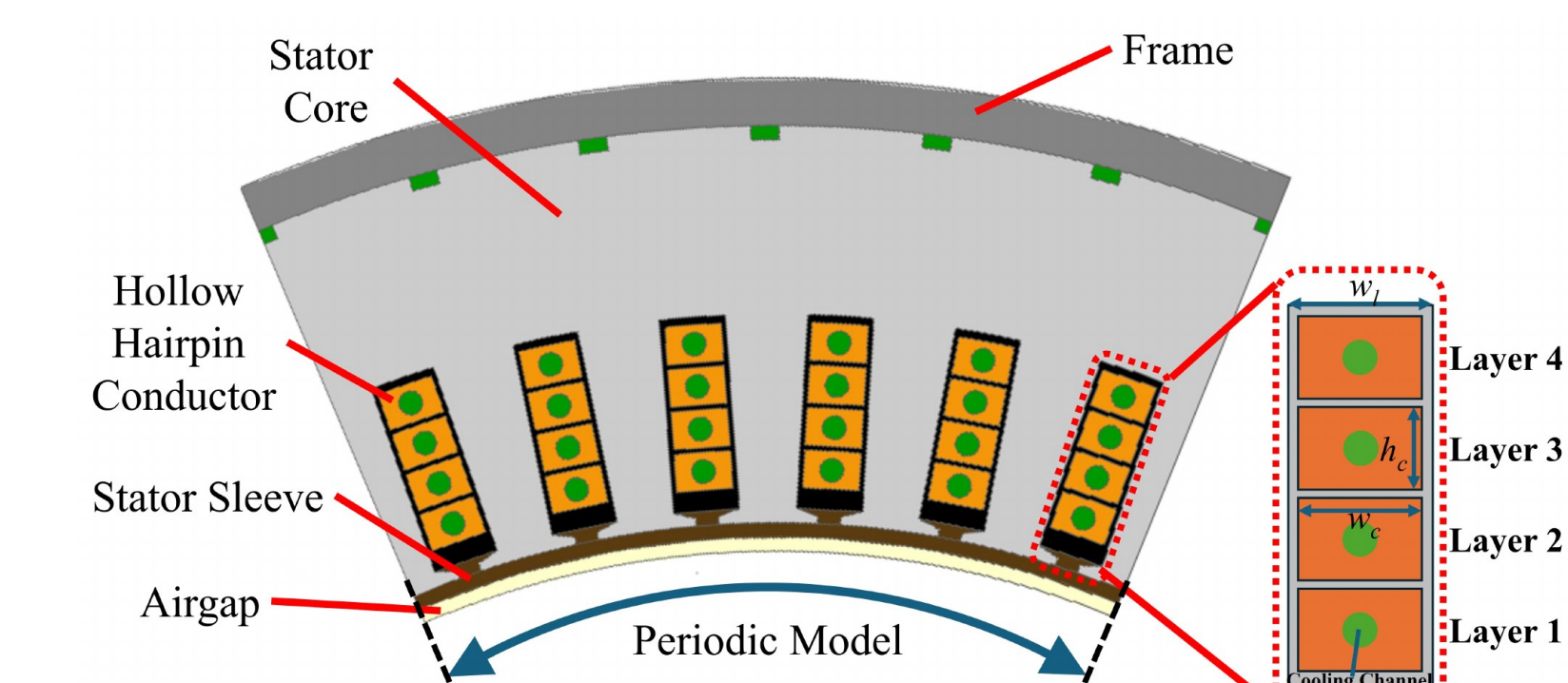
Based on the authors' ICEM 2026 and ECCE 2026 conference papers

Motivation

- Electrified traction pushes motors to higher speeds and frequencies — hairpin windings are the standard thanks to high slot fill and automated manufacturing.
- Skin and proximity effects concentrate AC losses in the conductors nearest the slot opening: severe local hot spots limit the continuous power rating.
- Water jackets remove heat only after it has crossed slot insulation and stator core — a long, high-resistance thermal path.

Direct oil cooling through hollow conductors

- Dielectric oil flows through a channel inside each hairpin conductor: heat is extracted directly at the source, hot spots are suppressed and the critical end-winding region is cooled too.
- Direct liquid cooling can sustain up to 277 % higher continuous current than a water jacket at the same winding temperature limit.

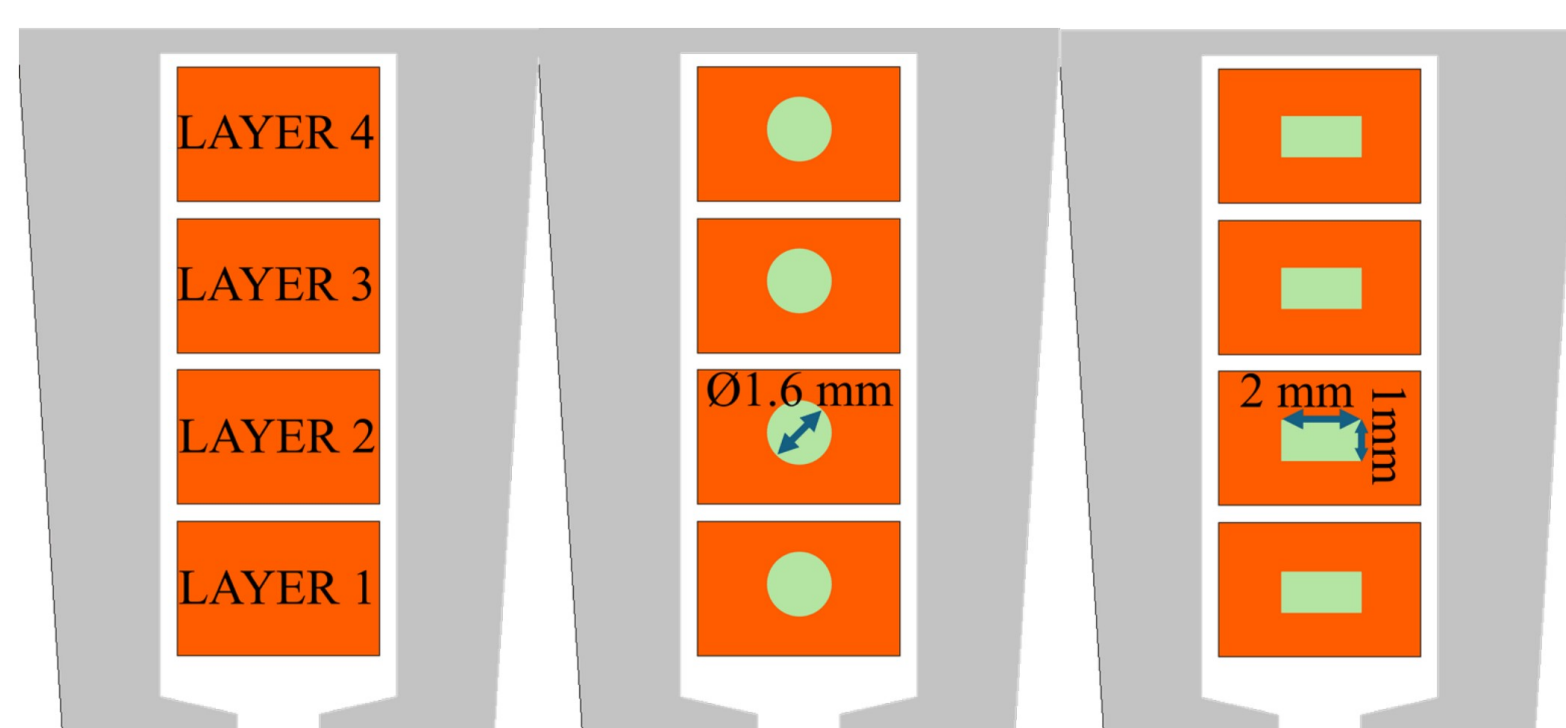


Reference machine: 300 kW hairpin-winding PMSM, modeled as a 1/6 periodic sector. Every conductor carries an internal cooling channel.

Reference machine & fair comparison

Rated power / speed	300 kW / 2000 rpm
Stator slots × layers	48 × 4 hollow conductors
Conductor cross-section	6 mm × 4 mm
Rated current / frequency	350 A rms / 266.7 Hz
Coolant (dielectric oil)	65 °C inlet, 10 L/min total
Flow regime	Laminar (Re ≤ 113)

- Fair-comparison rule: every channel keeps the same flow area (2 mm²) → same mean oil velocity and same copper removed — the pure shape effect is isolated.

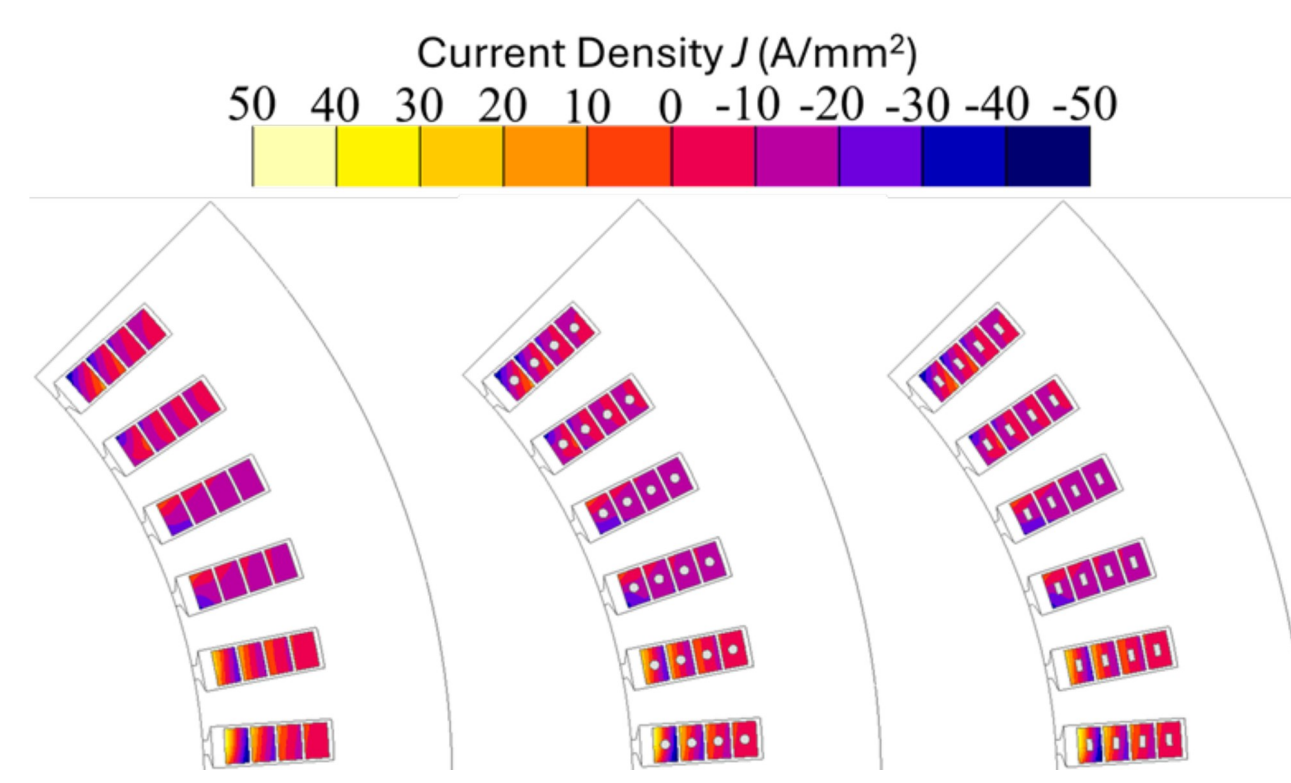


Compared slot topologies: solid baseline, round channel Ø 1.6 mm, and rectangular channel 1 mm × 2 mm.

Can the internal channel be shaped freely for cooling — without an electromagnetic penalty?

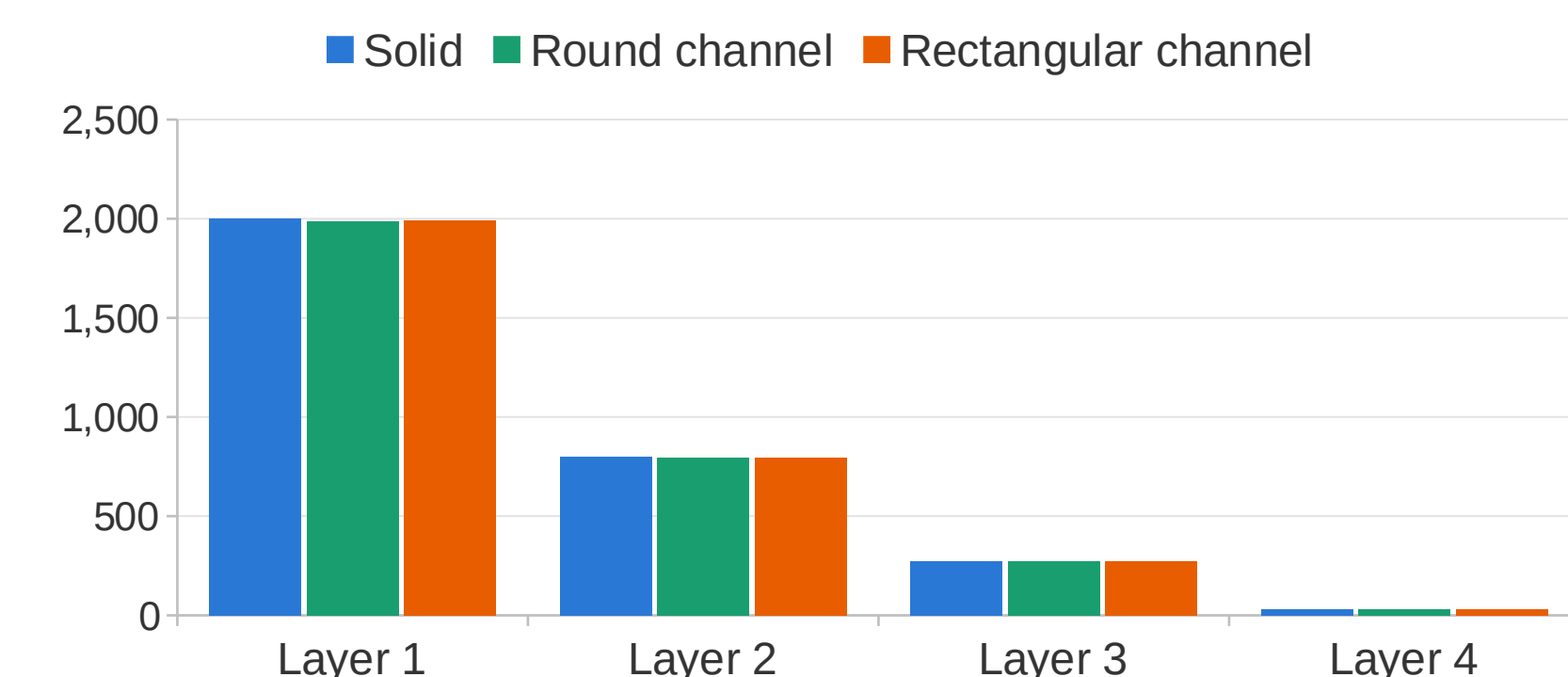
Electromagnetic analysis

- 2D FEA of all three topologies at the rated worst case: 350 A rms, 266.7 Hz.
- High-frequency current crowds at the conductor periphery — the centre acts as a low-current “dead zone”, exactly where the cooling channel sits.



Current density at rated frequency: solid, round and rectangular conductors (left → right). The channel region carries almost no current.

High-frequency AC losses per layer (W)



Layer 1 lies next to the airgap and dominates the loss — yet its AC losses are shape-independent.

< 0.25 %

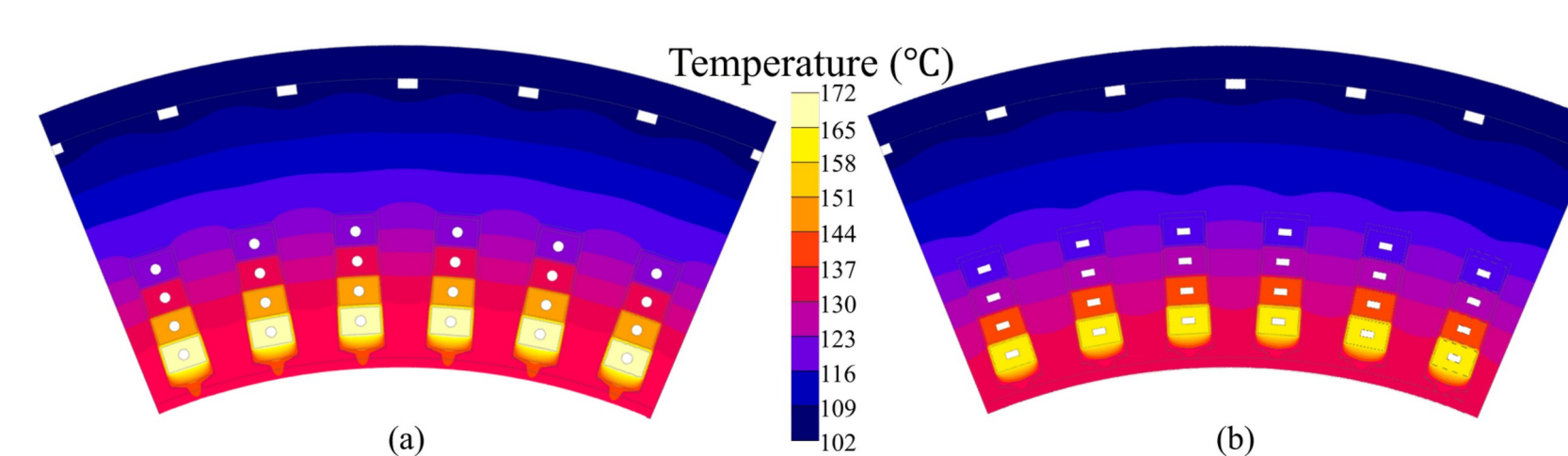
AC-loss difference between round and rectangular channels in the critical airgap-side layer — and only +2.4 % total loss for hollow vs solid conductors.

Thermal & hydraulic modelling

- Equal area ⇒ equal mean velocity: the shapes differ only in wetted perimeter P and hydraulic diameter $D_h = 4A/P$.
- Laminar flow ⇒ $h = Nu \cdot k / D_h$ — a larger wetted perimeter directly buys heat-transfer surface.

	Round	Rectangular
Wetted perimeter	5 mm	6 mm (+20 %)
Hydraulic diameter D_h	1.60 mm	1.33 mm
Nusselt number Nu	4.36	4.12
Heat-transfer coeff. h	≈ 350 W/(m ² K)	≈ 400 W/(m ² K)

Steady-state thermal FEA



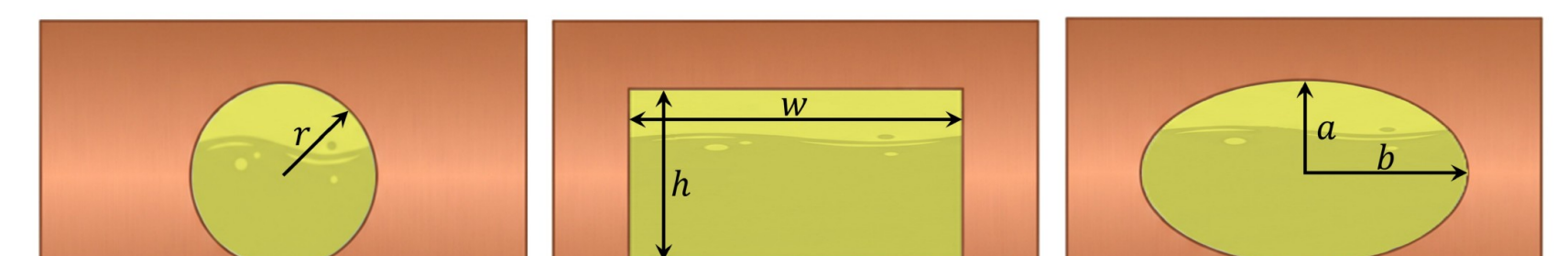
Temperature field at rated load — (a) round vs (b) rectangular channels. 2D FE slot model with layer-resolved losses and convection boundaries; the hot spot sits in Layer 1 next to the airgap.

−10 °C

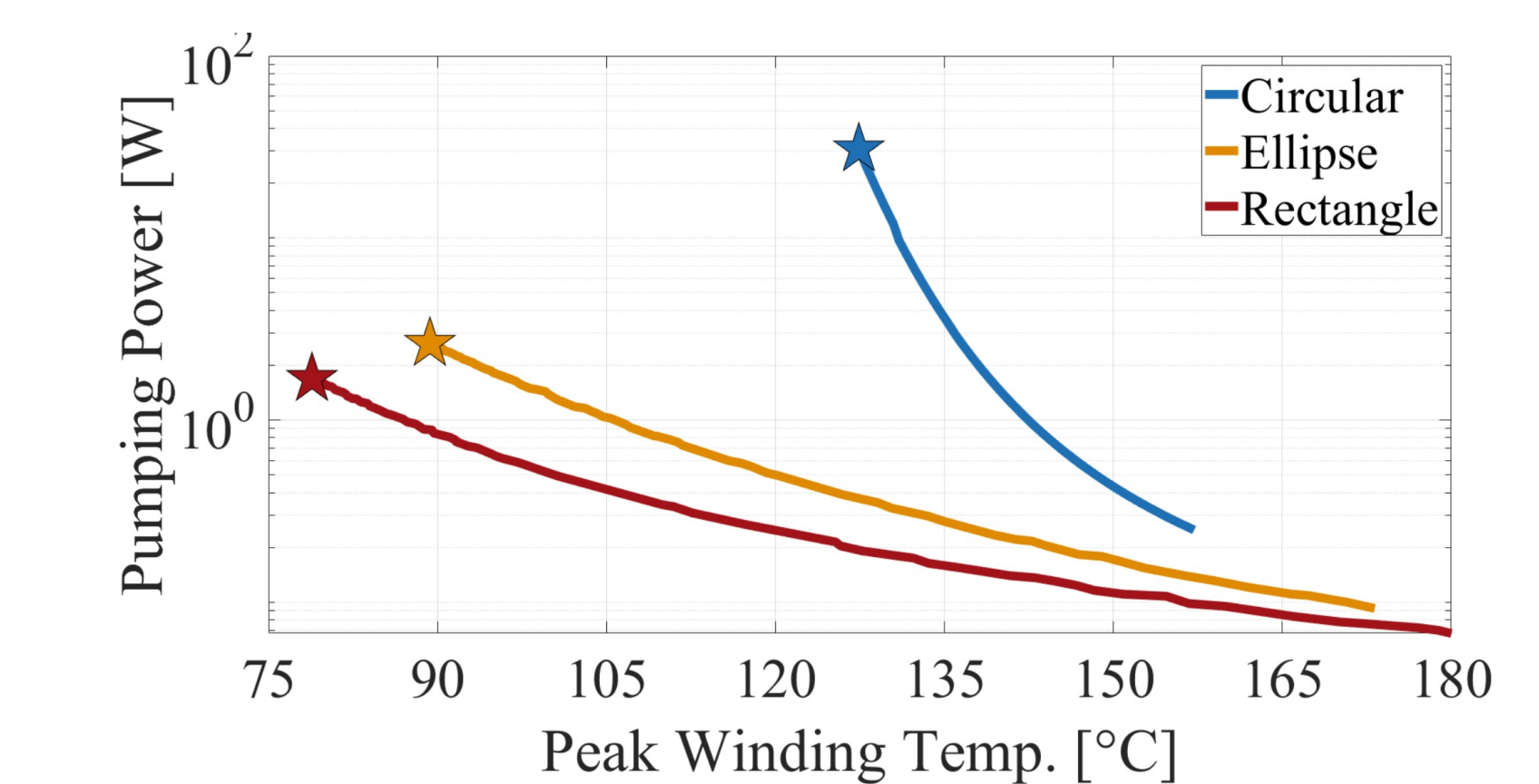
peak winding temperature: 172 °C (round) → 162 °C (rectangular) at the same 10 L/min oil flow — from the channel shape alone.

Channel-shape optimization

- Enlarging the channel adds wetted surface but removes copper (higher DC loss); shrinking it raises the pressure drop steeply — peak temperature and pumping power genuinely compete.
- Surrogate-based optimization: 128 thermal-FE samples per shape → surrogate model → multi-objective GA → Pareto front, with pumping power in closed form.
- Manufacturability constraints: wall thickness ≥ 0.75 mm, channel opening ≥ 0.75 mm.



Candidate shapes within the same conductor: circular, rectangular, elliptical.

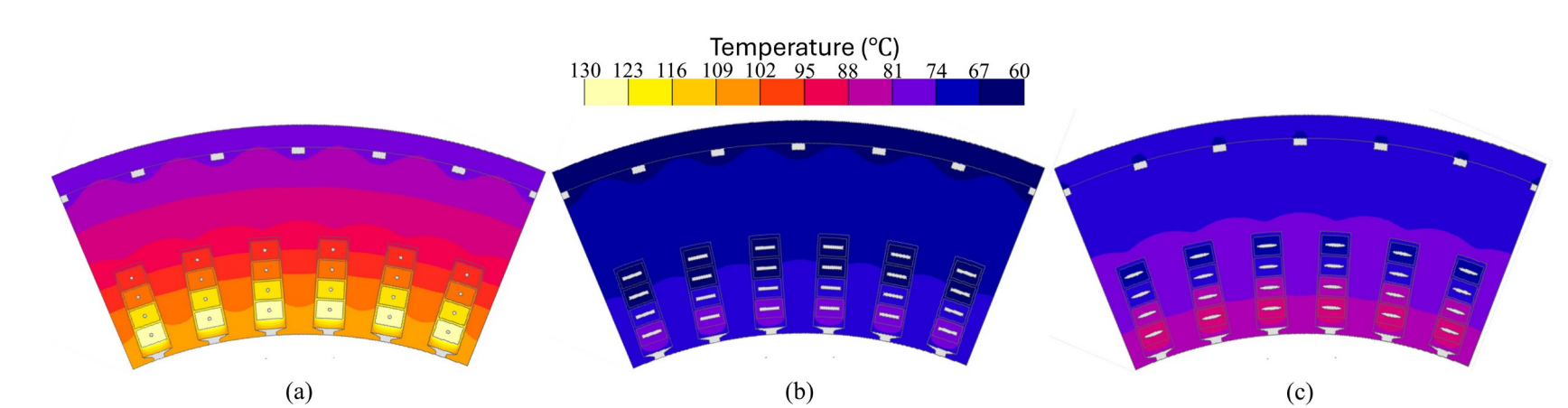


Pareto fronts (log pumping-power scale); stars mark minimum-temperature designs. The rectangular channel dominates the entire trade-off range.

Shape	Optimal channel	Peak T	Pump power
Circular	Ø 0.75 mm	127.4 °C	34.8 W
Elliptical	4.5 × 0.75 mm	89.3 °C	2.98 W
Rectangular	4.5 × 0.75 mm	78.8 °C	1.92 W

18 ×

less pumping power and −48.6 °C: the optimized rectangular channel (78.8 °C at 1.92 W) versus the optimized circular channel (127.4 °C at 34.8 W).



Temperature fields at each shape's minimum-temperature design: (a) circular, (b) rectangular, (c) elliptical. All optima are wide, shallow channels.

Conclusions

- The channel interior is electromagnetically “dead” — AC-loss penalty < 0.25 %: the channel can be shaped purely for cooling.
- Wetted perimeter within the fixed conductor envelope sets the thermal ranking: rectangle > ellipse >> circle.
- An equal-area swap from round to rectangular already buys −10 °C peak temperature.
- The optimized 4.5 × 0.75 mm rectangular channel reaches 78.8 °C at only 1.92 W pumping power.
- Next: manufacturing of the optimized hollow profiles and experimental validation on a motorette.

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